

ALFAA42554

## Sulfuric acid, 15% v/v aqueous solution

### SECTION 1. IDENTIFICATION OF THE SUBSTANCE/MIXTURE AND OF THE COMPANY/UNDERTAKING

产品说明:  
Product Description: 硫酸, 15% v/v 水溶液  
Sulfuric acid, 15% v/v aqueous solution

Cat No. : 42554

Supplier Avocado Research Chemicals Ltd.  
(Part of Thermo Fisher Scientific)  
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Emergency Number **US**:001-201-796-7100 / **Europe**: +32 14 57 52 99  
**CHEMTREC** Tel. No. **US**:001-800-424-9300 / **Europe**:001-703-527-3887

E-mail address begel.sdsdesk@thermofisher.com

Recommended Use Laboratory chemicals.  
Uses advised against No Information available

### SECTION 2. HAZARD IDENTIFICATION

**Physical State**  
Liquid

**Appearance**  
Colorless

**Odor**  
No information available

**Emergency Overview**  
Causes severe skin burns and eye damage.

#### Classification of the substance or mixture

|                                   |              |
|-----------------------------------|--------------|
| Skin Corrosion/Irritation         | Category 1 A |
| Serious Eye Damage/Eye Irritation | Category 1   |

#### Label Elements



Signal Word

**Danger**

#### Hazard Statements

H314 - Causes severe skin burns and eye damage

#### Precautionary Statements

# SAFETY DATA SHEET

## Sulfuric acid, 15% v/v aqueous solution

**Prevention**

P260 - Do not breathe dust/fume/gas/mist/vapors/spray  
P264 - Wash face, hands and any exposed skin thoroughly after handling  
P271 - Use only outdoors or in a well-ventilated area  
P280 - Wear protective gloves/protective clothing/eye protection/face protection

**Response**

P301 + P330 + P331 - IF SWALLOWED: rinse mouth. Do NOT induce vomiting  
P303 + P361 + P353 - IF ON SKIN (or hair): Take off immediately all contaminated clothing. Rinse skin with water or shower  
P304 + P340 - IF INHALED: Remove person to fresh air and keep comfortable for breathing  
P305 + P351 + P338 - IF IN EYES: Rinse cautiously with water for several minutes. Remove contact lenses, if present and easy to do. Continue rinsing  
P310 - Immediately call a POISON CENTER or doctor  
P362 + P364 - Take off contaminated clothing and wash it before reuse

**Storage**

P403 + P233 - Store in a well-ventilated place. Keep container tightly closed

**Disposal**

P501 - Dispose of contents/ container to an approved waste disposal plant

**Physical and Chemical Hazards**

None identified.

**Health Hazards**

Corrosive. Causes skin and eye burns. Causes serious eye damage.

**Environmental hazards**

Contains no substances known to be hazardous to the environment or not degradable in waste water treatment plants. Will likely be mobile in the environment due to its water solubility. The product is water soluble, and may spread in water systems.

This product does not contain any known or suspected endocrine disruptors.

**SECTION 3. COMPOSITION/INFORMATION ON INGREDIENTS**

| Component     | CAS No    | Weight % |
|---------------|-----------|----------|
| Water         | 7732-18-5 | 50-85    |
| Sulfuric acid | 7664-93-9 | 15-50    |

**SECTION 4. FIRST AID MEASURES****Eye Contact**

Rinse immediately with plenty of water, also under the eyelids, for at least 15 minutes. Immediate medical attention is required.

**Skin Contact**

Wash off immediately with plenty of water for at least 15 minutes. Immediate medical attention is required.

**Inhalation**

Remove to fresh air. Do not use mouth-to-mouth method if victim ingested or inhaled the substance; give artificial respiration with the aid of a pocket mask equipped with a one-way valve or other proper respiratory medical device. Immediate medical attention is required. If not breathing, give artificial respiration.

**Ingestion**

Do NOT induce vomiting. Call a physician or poison control center immediately.

**Most important symptoms and effects**

Causes burns by all exposure routes. Ingestion causes severe swelling, severe damage to the delicate tissue and danger of perforation: Product is a corrosive material. Use of gastric lavage or emesis is contraindicated. Possible perforation of stomach or esophagus should be investigated

**Self-Protection of the First Aider**

Ensure that medical personnel are aware of the material(s) involved, take precautions to protect themselves and prevent spread of contamination.

# SAFETY DATA SHEET

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**Notes to Physician**

Treat symptomatically.

**SECTION 5. FIRE-FIGHTING MEASURES****Suitable Extinguishing Media**

Use extinguishing measures that are appropriate to local circumstances and the surrounding environment.

**Extinguishing media which must not be used for safety reasons**

No information available.

**Specific Hazards Arising from the Chemical**

Thermal decomposition can lead to release of irritating gases and vapors.

**Protective Equipment and Precautions for Firefighters**

As in any fire, wear self-contained breathing apparatus pressure-demand, MSHA/NIOSH (approved or equivalent) and full protective gear. Thermal decomposition can lead to release of irritating gases and vapors.

**SECTION 6. ACCIDENTAL RELEASE MEASURES****Personal Precautions**

Ensure adequate ventilation. Avoid contact with skin, eyes or clothing. Use personal protective equipment as required. Keep people away from and upwind of spill/leak. Evacuate personnel to safe areas.

**Environmental Precautions**

Should not be released into the environment. See Section 12 for additional Ecological Information.

**Methods for Containment and Clean Up**

Soak up with inert absorbent material. Keep in suitable, closed containers for disposal.

Refer to protective measures listed in Sections 8 and 13.

**SECTION 7. HANDLING AND STORAGE****Handling**

Wear personal protective equipment/face protection. Do not get in eyes, on skin, or on clothing. Do not breathe (dust, vapor, mist, gas). Do not ingest. If swallowed then seek immediate medical assistance.

**Storage**

Keep containers tightly closed in a dry, cool and well-ventilated place. Corrosives area.

**Specific Use(s)**

Use in laboratories

**SECTION 8. EXPOSURE CONTROLS/PERSONAL PROTECTION****Control Parameters**

| Component     | China                                                 | Taiwan                   | Thailand                 | Hong Kong                                             |
|---------------|-------------------------------------------------------|--------------------------|--------------------------|-------------------------------------------------------|
| Sulfuric acid | TWA: 1 mg/m <sup>3</sup><br>STEL: 2 mg/m <sup>3</sup> | TWA: 1 mg/m <sup>3</sup> | TWA: 1 mg/m <sup>3</sup> | TWA: 1 mg/m <sup>3</sup><br>STEL: 3 mg/m <sup>3</sup> |

| Component     | ACGIH TLV                  | OSHA PEL                                                       | NIOSH                                                  | The United Kingdom                                                      | European Union                   |
|---------------|----------------------------|----------------------------------------------------------------|--------------------------------------------------------|-------------------------------------------------------------------------|----------------------------------|
| Sulfuric acid | TWA: 0.2 mg/m <sup>3</sup> | (Vacated) TWA: 1 mg/m <sup>3</sup><br>TWA: 1 mg/m <sup>3</sup> | IDLH: 15 mg/m <sup>3</sup><br>TWA: 1 mg/m <sup>3</sup> | STEL: 0.15 mg/m <sup>3</sup> 15 min<br>TWA: 0.05 mg/m <sup>3</sup> 8 hr | TWA: 0.05 mg/m <sup>3</sup> (8h) |

Legend

# SAFETY DATA SHEET

## Sulfuric acid, 15% v/v aqueous solution

ACGIH - American Conference of Governmental Industrial Hygienists  
 OSHA - Occupational Safety and Health Administration  
 NIOSH: NIOSH - National Institute for Occupational Safety and Health

### Exposure Controls

#### Engineering Measures

Ensure that eyewash stations and safety showers are close to the workstation location. Wherever possible, engineering control measures such as the isolation or enclosure of the process, the introduction of process or equipment changes to minimise release or contact, and the use of properly designed ventilation systems, should be adopted to control hazardous materials at source.

#### Personal protective equipment

**Eye Protection** Goggles (European standard - EN 166)

**Hand Protection** Protective gloves

| Glove material | Breakthrough time | Glove thickness | EU standard | Glove comments        |
|----------------|-------------------|-----------------|-------------|-----------------------|
| Butyl rubber   | > = 120 minutes   | 0.5 mm          | EN 374      | (minimum requirement) |
| Viton (R)      | > 480 minutes     | 0.4 mm          |             |                       |

Inspect gloves before use.

Please observe the instructions regarding permeability and breakthrough time which are provided by the supplier of the gloves. (Refer to manufacturer/supplier for information)

Ensure gloves are suitable for the task: Chemical compatibility, Dexterity, Operational conditions, User susceptibility, e.g. sensitisation effects, also take into consideration the specific local conditions under which the product is used, such as the danger of cuts, abrasion.

Remove gloves with care avoiding skin contamination.

**Skin and body protection** Wear appropriate protective gloves and clothing to prevent skin exposure

**Respiratory Protection** When workers are facing concentrations above the exposure limit they must use appropriate certified respirators.  
 To protect the wearer, respiratory protective equipment must be the correct fit and be used and maintained properly

**Large scale/emergency use** Use a NIOSH/MSHA or European Standard EN 136 approved respirator if exposure limits are exceeded or if irritation or other symptoms are experienced  
**Recommended Filter type:** Particulates filter conforming to EN 143

**Small scale/Laboratory use** Use a NIOSH/MSHA or European Standard EN 149:2001 approved respirator if exposure limits are exceeded or if irritation or other symptoms are experienced.  
**Recommended half mask:-** Valve filtering: EN405; or; Half mask: EN140; plus filter, EN 141  
 When RPE is used a face piece Fit Test should be conducted

**Hygiene Measures** Handle in accordance with good industrial hygiene and safety practice.

**Environmental exposure controls** No information available.

## SECTION 9. PHYSICAL AND CHEMICAL PROPERTIES

|                            |                          |                                          |
|----------------------------|--------------------------|------------------------------------------|
| <b>Appearance</b>          | Colorless                |                                          |
| <b>Physical State</b>      | Liquid                   |                                          |
| <b>Odor</b>                | No information available |                                          |
| <b>Odor Threshold</b>      | No data available        |                                          |
| <b>pH</b>                  | ~ 1 @ 20°C               | 5 g/l aq.sol                             |
| <b>Melting Point/Range</b> | No data available        |                                          |
| <b>Softening Point</b>     | No data available        |                                          |
| <b>Boiling Point/Range</b> | approx 120 °C            |                                          |
| <b>Flash Point</b>         | Not applicable           | <b>Method -</b> No information available |

# SAFETY DATA SHEET

## Sulfuric acid, 15% v/v aqueous solution

|                                         |                          |             |
|-----------------------------------------|--------------------------|-------------|
| Evaporation Rate                        | No data available        |             |
| Flammability (solid,gas)                | Not applicable           | Liquid      |
| Explosion Limits                        | No data available        |             |
| Vapor Pressure                          | No data available        |             |
| Vapor Density                           | No data available        | (Air = 1.0) |
| Specific Gravity / Density              | 1.10-1.30                |             |
| Bulk Density                            | Not applicable           | Liquid      |
| Water Solubility                        | Soluble                  |             |
| Solubility in other solvents            | No information available |             |
| Partition Coefficient (n-octanol/water) |                          |             |
| Autoignition Temperature                | Not applicable           |             |
| Decomposition Temperature               | No data available        |             |
| Viscosity                               | No data available        |             |
| Explosive Properties                    | No information available |             |
| Oxidizing Properties                    | No information available |             |

### SECTION 10. STABILITY AND REACTIVITY

|                                  |                                                        |
|----------------------------------|--------------------------------------------------------|
| Stability                        | Stable under normal conditions.                        |
| Hazardous Reactions              | Contact with metals may evolve flammable hydrogen gas. |
| Hazardous Polymerization         | Hazardous polymerization does not occur.               |
| Conditions to Avoid              | Incompatible products. Excess heat.                    |
| Materials to avoid               | Strong bases. Metals. Reducing Agent.                  |
| Hazardous Decomposition Products | Sulfur oxides.                                         |

### SECTION 11. TOXICOLOGICAL INFORMATION

#### Product Information

(a) acute toxicity;  
Toxicology data for the components

| Component     | LD50 Oral          | LD50 Dermal | LC50 Inhalation               |
|---------------|--------------------|-------------|-------------------------------|
| Water         | -                  | -           | -                             |
| Sulfuric acid | 2140 mg/kg ( Rat ) |             | LC50 = 0.375 mg/L ( Rat ) 4 h |

(b) skin corrosion/irritation; Category 1 A

(c) serious eye damage/irritation; Category 1

(d) respiratory or skin sensitization;

|             |                   |
|-------------|-------------------|
| Respiratory | No data available |
| Skin        | No data available |

(e) germ cell mutagenicity; No data available

(f) carcinogenicity; No data available

The table below indicates whether each agency has listed any ingredient as a carcinogen

# SAFETY DATA SHEET

## Sulfuric acid, 15% v/v aqueous solution

| Component     | EU | UK | Germany | IARC    |
|---------------|----|----|---------|---------|
| Sulfuric acid |    |    |         | Group 1 |

(g) reproductive toxicity; No data available

(h) STOT-single exposure; No data available

(i) STOT-repeated exposure; No data available

Target Organs No information available.

(j) aspiration hazard; No data available

**Symptoms / effects, both acute and delayed** Ingestion causes severe swelling, severe damage to the delicate tissue and danger of perforation: Product is a corrosive material. Use of gastric lavage or emesis is contraindicated. Possible perforation of stomach or esophagus should be investigated

### SECTION 12. ECOLOGICAL INFORMATION

**Ecotoxicity effects** Do not empty into drains. .

| Component     | Freshwater Fish                                  | Water Flea        | Freshwater Algae | Microtox |
|---------------|--------------------------------------------------|-------------------|------------------|----------|
| Sulfuric acid | LC50: > 500 mg/L, 96h static (Brachydanio rerio) | EC50: 29 mg/L/24h | -                | -        |

**Persistence and Degradability**  
**Persistence** Soluble in water, Persistence is unlikely, based on information available.

**Bioaccumulative Potential** Bioaccumulation is unlikely

**Mobility in soil** The product is water soluble, and may spread in water systems Will likely be mobile in the environment due to its water solubility Highly mobile in soils

**Endocrine Disruptor Information** This product does not contain any known or suspected endocrine disruptors  
**Persistent Organic Pollutant** This product does not contain any known or suspected substance  
**Ozone Depletion Potential** This product does not contain any known or suspected substance

### SECTION 13. DISPOSAL CONSIDERATIONS

**Waste from Residues/Unused Products** Waste is classified as hazardous. Dispose of in accordance with the European Directives on waste and hazardous waste. Dispose of in accordance with local regulations.

**Contaminated Packaging** Dispose of this container to hazardous or special waste collection point.

**Other Information** Waste codes should be assigned by the user based on the application for which the product was used. Do not empty into drains. Do not flush to sewer. Large amounts will affect pH and harm aquatic organisms. Solutions with low pH-value must be neutralized before discharge.

### SECTION 14. TRANSPORT INFORMATION

# SAFETY DATA SHEET

## Sulfuric acid, 15% v/v aqueous solution

### Road and Rail Transport

UN-No UN2796  
Proper Shipping Name Sulphuric acid  
Hazard Class 8  
Packing Group II

### IMDG/IMO

UN-No UN2796  
Proper Shipping Name Sulphuric acid  
Hazard Class 8  
Packing Group II

### IATA

UN-No UN2796  
Proper Shipping Name Sulphuric acid  
Hazard Class 8  
Packing Group II

Special Precautions for User No special precautions required

## SECTION 15. REGULATORY INFORMATION

### International Inventories

X = listed, China (IECSC), Europe (EINECS/ELINCS/NLP), U.S.A. (TSCA), Canada (DSL/NDSL), Philippines (PICCS), Japan (ENCS), Japan (ISHL), Australia (AICS), Korea (KECL).

| Component     | The Inventory of Hazardous Chemicals (2015 Edition) | List of dangerous goods GB 12268 - 2012 | TCSI | IECSC | EINECS    | TSCA | DSL | PICCS | ENCS | ISHL | AICS | KECL     |
|---------------|-----------------------------------------------------|-----------------------------------------|------|-------|-----------|------|-----|-------|------|------|------|----------|
| Water         | -                                                   | -                                       | X    | X     | 231-791-2 | X    | X   | X     | X    |      | X    | KE-35400 |
| Sulfuric acid | X                                                   | X                                       | X    | X     | 231-639-5 | X    | X   | X     | X    | X    | X    | KE-32570 |

### National Regulations

## SECTION 16. OTHER INFORMATION

Prepared By Health, Safety and Environmental Department  
Creation Date 23-Jun-2009  
Revision Date 08-May-2024  
Revision Summary New emergency telephone response service provider.

### Training Advice

Chemical hazard awareness training, incorporating labelling, Safety Data Sheets (SDS), Personal Protective Equipment (PPE) and hygiene.

Use of personal protective equipment, covering appropriate selection, compatibility, breakthrough thresholds, care, maintenance, fit and standards.

First aid for chemical exposure, including the use of eye wash and safety showers.

Chemical incident response training.

### Legend

# SAFETY DATA SHEET

## Sulfuric acid, 15% v/v aqueous solution

**CAS** - Chemical Abstracts Service

**EINECS/ELINCS** - European Inventory of Existing Commercial Chemical Substances/EU List of Notified Chemical Substances

**PICCS** - Philippines Inventory of Chemicals and Chemical Substances

**IECSC** - Chinese Inventory of Existing Chemical Substances

**KECL** - Korean Existing and Evaluated Chemical Substances

**TSCA** - United States Toxic Substances Control Act Section 8(b) Inventory

**DSL/NDL** - Canadian Domestic Substances List/Non-Domestic Substances List

**ENCS** - Japanese Existing and New Chemical Substances

**AICS** - Australian Inventory of Chemical Substances

**NZIoC** - New Zealand Inventory of Chemicals

**WEL** - Workplace Exposure Limit

**ACGIH** - American Conference of Governmental Industrial Hygienists

**DNEL** - Derived No Effect Level

**RPE** - Respiratory Protective Equipment

**LC50** - Lethal Concentration 50%

**NOEC** - No Observed Effect Concentration

**PBT** - Persistent, Bioaccumulative, Toxic

**TWA** - Time Weighted Average

**IARC** - International Agency for Research on Cancer

**PNEC** - Predicted No Effect Concentration

**LD50** - Lethal Dose 50%

**EC50** - Effective Concentration 50%

**POW** - Partition coefficient Octanol:Water

**vPvB** - very Persistent, very Bioaccumulative

**ICAO/IATA** - International Civil Aviation Organization/International Air Transport Association

**ADR** - European Agreement Concerning the International Carriage of Dangerous Goods by Road

**OECD** - Organisation for Economic Co-operation and Development

**BCF** - Bioconcentration factor

**IMO/IMDG** - International Maritime Organization/International Maritime Dangerous Goods Code

**MARPOL** - International Convention for the Prevention of Pollution from Ships

**ATE** - Acute Toxicity Estimate

**VOC** - (Volatile Organic Compound)

### Key literature references and sources for data

<https://echa.europa.eu/information-on-chemicals>

Suppliers safety data sheet, Chemadvisor - LOLI, Merck index, RTECS

**Physical hazards**

On basis of test data

**Health Hazards**

Calculation method

**Environmental hazards**

Calculation method

### Disclaimer

The information provided in this Safety Data Sheet is correct to the best of our knowledge, information and belief at the date of its publication. The information given is designed only as a guidance for safe handling, use, processing, storage, transportation, disposal and release and is not to be considered a warranty or quality specification. The information relates only to the specific material designated and may not be valid for such material used in combination with any other materials or in any process, unless specified in the text

**End of Safety Data Sheet**